

MDCORA Technologies Pvt. Ltd.

PROJECT ASSIGNMENT 2

1. Project Title

AI/ML Marketing Campaign Intelligence

2. Project Domain

Artificial Intelligence & Machine Learning

3. Project Level

Intermediate

4. Project Duration

5-7 Days

5. Project Description

The **AI/ML Marketing Campaign Intelligence** project focuses on developing a Machine Learning-based system that analyses historical marketing campaign data to identify performance patterns, predict campaign outcomes, segment audiences, and detect unusual campaign behaviour.

You will work on the complete Machine Learning lifecycle, starting from data collection and preprocessing to exploratory data analysis, feature engineering, model development, model evaluation, optimization, and final prediction. The project will use marketing metrics such as impressions, clicks, CTR, CPC, conversions, conversion rate, advertising spend, revenue, and ROAS to develop meaningful Machine Learning solutions.

The system will include multiple ML components such as **campaign performance prediction, campaign classification, audience segmentation, and anomaly detection**. The objective is to transform raw marketing data into useful predictive insights that can support data-driven marketing decisions.

6. Project Objective

The main objectives of the project are:

- Analyze historical marketing campaign data.
- Identify factors affecting campaign performance.

- Create meaningful marketing KPIs and ML features.
 - Predict campaign performance.
 - Classify campaigns based on performance.
 - Segment customers/audiences using Machine Learning.
 - Detect unusual campaign behaviour.
 - Compare and optimize different ML models.
 - Generate meaningful marketing insights from ML results.
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7. Project Workflow

Marketing Dataset → Data Cleaning → EDA → KPI Calculation → Feature Engineering → ML Model Development → Model Evaluation → Model Optimization → Prediction → Segmentation → Anomaly Detection → Final ML Report

8. Step-by-Step Work

Phase 1 – Dataset Collection & Understanding

You will collect or work with a marketing campaign dataset containing information such as:

- Campaign ID
- Campaign Name
- Marketing Channel
- Campaign Type
- Budget
- Impressions
- Reach
- Clicks
- Spend
- Conversions
- Revenue
- Campaign Duration

- Audience Information

Expected Output

- Raw dataset
 - Data dictionary
 - Initial dataset analysis
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Phase 2 – Data Cleaning & Preprocessing

You will prepare the dataset for Machine Learning.

Work will include:

- Handling missing values
- Removing duplicate records
- Correcting data types
- Handling inconsistent values
- Detecting outliers
- Encoding categorical variables
- Scaling numerical features where required

Expected Output

Cleaned and ML-ready dataset

Phase 3 – Exploratory Data Analysis

You will analyze marketing data to identify important patterns and relationships.

Analysis will include:

- Campaign performance
- Channel performance
- Budget vs revenue
- Spend vs conversions
- CTR analysis
- CPC analysis

- Conversion rate analysis
- ROAS analysis
- Correlation analysis

Expected Output

- EDA notebook
- Charts and visualizations
- EDA findings report

9. Marketing KPI & Feature Engineering

You will create meaningful marketing metrics from raw data.

Important KPIs

CTR

$\text{CTR} = \text{Clicks} / \text{Impressions} \times 100$

Conversion Rate

$\text{Conversion Rate} = \text{Conversions} / \text{Clicks} \times 100$

CPC

$\text{CPC} = \text{Spend} / \text{Clicks}$

CPA

$\text{CPA} = \text{Spend} / \text{Conversions}$

ROAS

$\text{ROAS} = \text{Revenue} / \text{Spend}$

The intern will also create additional ML features based on campaign and customer behavior.

Expected Output

Feature-engineered dataset

10. Machine Learning Model 1 – Campaign Performance Prediction

The first ML task will be to predict campaign performance using historical marketing data.

Possible prediction targets:

- Revenue
- Conversions
- ROAS
- Clicks

Algorithms

- Linear Regression
- Random Forest Regression
- XG Boost Regression

Evaluation Metrics

- MAE
- MSE
- RMSE
- R^2 Score

Expected Output

Campaign Performance Prediction Model

11. Machine Learning Model 2 – Campaign Classification

You will classify marketing campaigns according to their historical performance.

Example:

- High Performing
- Medium Performing
- Low Performing

The performance labels should be defined using appropriate business rules and validated against the dataset.

Algorithms

- Logistic Regression
- Decision Tree

- Random Forest
- XGBoost

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

Expected Output

Campaign Classification Model

12. Machine Learning Model 3 – Audience Segmentation

You will use unsupervised Machine Learning to identify groups of customers/audiences based on their marketing behaviour.

Technique

K-Means Clustering

Possible segments:

- Highly Engaged Audience
- High Conversion Audience
- Low Engagement Audience
- Potential Customers

You will analyse and interpret each cluster instead of simply generating cluster numbers.

Evaluation

- Silhouette Score
- Cluster analysis

Expected Output

Audience Segmentation Model

13. Machine Learning Model 4 – Marketing Anomaly Detection

You will develop an approach to identify unusual marketing campaign behavior.

Examples:

- Sudden increase in CPC
- Significant CTR decrease
- Unexpected conversion drop
- Abnormal advertising spend
- Unusual ROAS

Possible Techniques

- Isolation Forest
- Statistical Anomaly Detection

Expected Output

Marketing Campaign Anomaly Detection Model

14. Model Comparison & Optimization

You will compare multiple algorithms and select suitable models based on evaluation results.

Work will include:

- Model comparison
- Cross-validation
- Hyperparameter tuning
- Feature selection
- Feature importance analysis
- Model optimization

Expected Output

Final Selected ML Models + Model Comparison Report

15. Prediction Pipeline

You will create a complete ML prediction workflow:

Input Marketing Data



Data Preprocessing



Feature Engineering



Trained ML Model



Prediction



Marketing Insight

The pipeline should be structured so that new marketing data can be processed using the trained models.

16. Technology Stack

Programming

Python

Data Processing

Pandas, NumPy

Machine Learning

Scikit-learn, XGBoost

Data Visualization

Matplotlib, Seaborn

Development Environment

Jupyter Notebook / VS Code

Version Control

Git / GitHub

Database

17. Expected Deliverables

At the end of the project, the intern should submit:

1. Raw marketing dataset
2. Data dictionary
3. Cleaned dataset
4. EDA notebook
5. EDA report
6. Feature engineering notebook/code
7. Campaign performance prediction model
8. Campaign classification model
9. Audience segmentation model
10. Anomaly detection model
11. Model evaluation report
12. Model comparison report
13. Feature importance analysis
14. Final ML prediction pipeline
15. Source code repository
16. Technical documentation
17. Final project presentation

18. Final Project Outcome

The final outcome will be an **AI/ML-based Marketing Campaign Intelligence system** capable of analyzing historical marketing data, predicting campaign performance, classifying campaigns, identifying audience segments, and detecting unusual campaign behavior.

The project will demonstrate the intern's ability to apply **Machine Learning to a real-world business problem** rather than only developing individual ML models.

19. Learning Outcomes

After completing the project, the intern will gain practical experience in:

- Python for Machine Learning
- Data preprocessing
- Exploratory Data Analysis
- Feature engineering
- Regression
- Classification
- Clustering
- Anomaly detection
- Model evaluation
- Hyperparameter tuning
- Feature importance
- ML pipeline development
- Business and marketing analytics
- Git/GitHub
- Technical documentation

20. Project Completion Criteria

The project will be considered complete when the intern has:

- ✓ Prepared and cleaned the dataset
- ✓ Completed EDA and KPI analysis
- ✓ Developed multiple ML models
- ✓ Evaluated and compared the models
- ✓ Completed audience segmentation
- ✓ Implemented anomaly detection
- ✓ Created a reusable ML prediction pipeline
- ✓ Documented the complete methodology and results
- ✓ Presented the final project and findings

Deployment is not included in the project scope.